

**Safety-Constrained Subscriber-Centric  
Broadband Infrastructure for  
Emerging Economies:  
Edge-Aware Slicing, Two-Timescale Control,  
and Nepal as an Empirical Testbed**

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**Simulation Stack:** NS-3, Mininet-WiFi, PyTorch Geometric

**Prototype Stack:** OpenWrt + FreeRADIUS + VLAN/tc + WireGuard/VXLAN

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# Abstract

Fixed broadband access in most emerging economies remains organised around a router-centric model in which a subscriber line, optical network terminal, and private Wi-Fi router define the practical boundary of internet service. This architecture fragments the access layer into isolated Wi-Fi islands, duplicates infrastructure in dense urban settings, and imposes high last-metre deployment costs in rural and mountainous regions. The central deficiency is not bandwidth alone; it is the absence of a subscriber-centric access fabric in which identity, authorisation, QoS entitlement, accounting, and regulatory traceability can follow an authenticated user across participating shared access points.

This proposal presents a safety-constrained subscriber-centric broadband architecture using Nepal as the primary empirical testbed while retaining applicability to emerging-economy access networks more generally. The design separates hard safety conditions - authentication validity, policy compliance, per-tenant slice isolation, and accounting consistency - from performance objectives. Learning-based optimisation is permitted only inside the state-dependent safe feasible action set.

The architecture integrates IEEE 802.1X/EAP-TLS and Passpoint-style identity provisioning, RADSEC-federated RADIUS authentication, policy-driven multi-tenant edge slicing through Linux `tc/HTB/fq_codel` and VLAN/VXLAN mechanisms, secure overlay transport through WireGuard or VXLAN, a two-timescale control model that separates commodity-edge fast-path enforcement from graph-neural-network macro-policy optimisation, and optional LEO/Starlink-class backhaul for rural Himalayan scenarios.

The revised mathematical model deliberately avoids pseudo-rigorous overclaiming. The finite implementation queue is modelled as a reflected/censored drop-tail process, while Large Deviations Theory is applied only to an auxiliary infinite-buffer workload. Effective-bandwidth root matching is stated using a supremum definition unless root existence is empirically or analytically justified. The CMDP is formulated consistently in cost-minimisation form with a state-dependent safe action set. LEO backoff equations use dimensionless normalised inputs and clamped control horizons. HMAC-based survivability targets are stated as empirical acceptance criteria, not cryptographic guarantees.

The expected output is a reproducible architecture and evaluation methodology for

secure, accountable, edge-aware, and optimisation-assisted shared broadband access, directly actionable for Nepal and generalisable to emerging-economy contexts with comparable access fragmentation, backhaul instability, and regulatory accountability constraints.

**Keywords:** subscriber-centric broadband; shared Wi-Fi; RADSEC; EAP-TLS; edge slicing; safe CMDP; effective bandwidth; two-timescale control; LEO backhaul; Nepal broadband; graph neural networks.

# Chapter 1

## Introduction

### 1.1 Background

Broadband expansion is usually measured through subscription counts, coverage ratios, average throughput, and affordability indices. These indicators are necessary, but they do not capture the architectural efficiency of the fixed broadband access edge. In residential, hostel, campus, and small-office deployments, access is typically bound to a private router controlled by one subscriber:

$$\text{one subscriber} \rightarrow \text{one line/ONT} \rightarrow \text{one router} \rightarrow \text{one isolated Wi-Fi island.} \quad (1.1)$$

This structure fragments the access layer. Neighbouring subscribers in the same building duplicate routers, SSIDs, cabling, power usage, spectrum occupation, and maintenance effort. In dense urban environments the dominant consequences are overlapping service sets, unmanaged co-channel interference, redundant access points, poor spectrum reuse, and absent roaming continuity. In rural and mountainous regions the same router-centric model manifests as high last-metre deployment cost, difficult maintenance, backhaul scarcity, and low infrastructure reuse.

Existing shared-access approaches solve only partial versions of this problem. Captive portals and public hotspots expand connectivity but lack cryptographic subscriber identity, ISP-grade per-user QoS enforcement, and reliable accounting. Enterprise Wi-Fi systems provide strong local management but do not generalise beyond one administrative domain. Passpoint and eduroam-style systems demonstrate that federated profile-based access can be operationalised at scale [5, 8], but they do not by themselves provide a complete model for broadband-package portability, edge slicing, regulatory traceability, and graph-optimised radio resource allocation. Cellular network slicing provides sophisticated control [9], but its infrastructure assumptions are not directly suited to low-cost Wi-Fi-based fixed broadband.



This research proposes a **subscriber-centric broadband model** in which the subscriber identity, rather than a local router password or SSID, becomes the primary control primitive for authentication, authorisation, QoS, slice enforcement, accounting, and auditability.

## 1.2 Nepal as Empirical Testbed and Generalisation Frame

Nepal is a strong empirical testbed because it combines dense urban broadband growth, rural and mountainous deployment difficulty, power and terrain constraints, dependence on middle-mile and upstream connectivity, and explicit regulatory interest in digital infrastructure [28, 29]. These conditions are not treated as isolated national peculiarities. They define a broader emerging-economy access class where fragmented last-metre infrastructure, uneven service quality, and accountability requirements coexist.

Four Nepal scenario classes structure the evaluation:

1. dense urban residential clusters, such as Kathmandu-class Wi-Fi-dense buildings;
2. semi-urban venue clusters, such as terai towns, campuses, hostels, and small commercial areas;
3. rural extension zones, such as hillside settlements with unstable terrestrial backhaul;
4. backhaul-constrained mountain settlements, where LEO satellite is evaluated as an optional resilience extension.

Starlink is therefore referenced only as a Starlink-class implementation of LEO satellite backhaul. The formal architecture remains vendor-neutral.

## 1.3 Research Problem

The central research problem is that fixed broadband access remains structurally router-centric. Subscriber entitlement is bound to a physical access device and local credentials rather than to a portable, authenticated, policy-governed identity. This creates three linked failures: duplicated access infrastructure, non-portable subscriber entitlement, and lack of a unified systems framework connecting identity, policy, slicing, telemetry, optimisation, and accounting.

**Research Problem Statement:** How can fixed broadband access be redesigned from a router-centric model into a subscriber-centric, identity-governed, multi-tenant edge access architecture that enables secure roaming

across participating access points while preserving ISP policy control, per-user QoS, traffic isolation, accounting continuity, and regulatory traceability in Nepal-like and broader emerging-economy environments?

## 1.4 Research Aim

The aim of this research is to design, implement, and evaluate a safety-constrained subscriber-centric shared broadband architecture that enables authenticated subscribers to access participating broadband edge nodes through secure identity federation, enforceable multi-tenant slicing, measurable QoS and accounting, and graph-assisted resource allocation within the resource constraints of commodity OpenWrt-class edge hardware.

## 1.5 Research Objectives

- O1.** Develop a subscriber-centric architecture separating trust, edge enforcement, backhaul selection, telemetry, optimisation, survivability, and audit functions.
- O2.** Implement a minimum viable prototype using OpenWrt, hostapd, FreeRADIUS, VLANs, Linux `tc`, VXLAN, and WireGuard while measuring edge CPU, RAM, interrupts, cryptographic overhead, tunnel overhead, and telemetry cost.
- O3.** Define a two-timescale control architecture in which AP-local fast-path enforcement handles packet-scale events while a graph-based optimiser updates macro policies at slower epochs.
- O4.** Formalise access allocation as a constrained optimisation problem where authentication, policy compliance, isolation, accounting, and edge-resource feasibility define the feasible action space.
- O5.** Evaluate graph-based allocation against random, greedy, static, and unconstrained-learning baselines without allowing violations of safety constraints.
- O6.** Scope physics-informed modelling correctly: indoor Wi-Fi uses log-distance/multi-wall models; rain and earth-space propagation models are reserved for LEO and long-range backhaul.
- O7.** Evaluate LEO/Starlink-class backhaul as an optional rural-resilience path rather than as a prerequisite.
- O8.** Model Nepal-specific survivability under power cuts, fibre disruption, and intermittent backhaul through token-cache degraded mode and signed offline accounting.

## 1.6 Research Questions

- RQ1.** How can subscriber identity be operationalised as the primary control primitive for authentication, authorisation, QoS enforcement, accounting, and slice isolation across shared broadband access points?
- RQ2.** To what extent can a commodity prototype based on OpenWrt, FreeRADIUS, VLAN/**tc**, WireGuard/VXLAN, and RADIUS accounting enforce secure multi-tenant broadband access while remaining within edge-resource feasibility bounds?
- RQ3.** Can graph-based macro-policy optimisation improve throughput, fairness, interference management, and SLA compliance compared with static, greedy, and unconstrained-learning baselines in Nepal-like dense and rural scenarios?

## 1.7 Research Contributions

The primary contributions are: (C1) a subscriber-centric access model; (C2) a safety-constrained multi-plane architecture; (C3) a safety-constrained resource-allocation formulation combining finite-buffer implementation loss, auxiliary effective-bandwidth overflow estimation, edge-aware graph telemetry, and state-dependent CMDP control without treating authentication failures, policy violations, slice leakage, accounting mismatch, or edge-resource infeasibility as optimisable soft penalties; (C4) a reproducible five-track evaluation methodology; and (C5) a Nepal-grounded deployment framework with optional LEO backhaul resilience.

# Chapter 2

## Literature Review and Theoretical Framework

### 2.1 Federated Authentication and Subscriber Identity

Network access control relies on Authentication, Authorisation, and Accounting mechanisms. RADIUS remains foundational; RADSEC secures RADIUS over TLS for inter-domain federation [2]. IEEE 802.1X defines port-based network access control [1], while EAP-TLS provides certificate-based mutual authentication [3]. The Network Access Identifier supports realm-based identity routing [4], and eduroam shows that federated roaming can operate at global scale [5].

These systems establish identity-based access but do not solve the whole subscriber-centric broadband problem. Authentication identifies the user, but does not by itself map broadband packages to shared AP slices, enforce traffic isolation, optimise radio resources, or maintain accounting consistency across constrained visited access points.

### 2.2 Wi-Fi Roaming, Hotspots, and Shared Access

Passpoint and OpenRoaming demonstrate automated profile-based Wi-Fi access and federation [7, 8]. Public hotspots and captive portals improve convenience but weaken cryptographic identity, per-user QoS, and accounting. Community networks show infrastructure sharing but usually lack standardised subscriber-policy enforcement and regulatory traceability.

The proposed system treats shared broadband as a multi-plane systems problem: identity federation, policy resolution, edge slicing, telemetry, accounting, and slow-timescale optimisation are evaluated as one reproducible architecture.

## 2.3 Edge Slicing and Secure Overlay Transport

Tenant separation can be implemented using VLANs, traffic-control queues, overlays, and programmable datapaths. VXLAN supports layer-2 overlays over layer-3 networks [6]. WireGuard provides lightweight encrypted tunnelling [12]. Linux `tc`, HTB, and `fq_codel` provide practical per-subscriber shaping on commodity Linux-based devices [13]. eBPF/XDP can support packet classification and telemetry on capable hardware [11], but it is not assumed to be feasible on all low-end APs.

## 2.4 LEO Backhaul and Rural Resilience

LEO satellite systems are relevant where terrestrial backhaul is unavailable or unstable. Measurement studies of Starlink-class systems show that LEO performance depends on routing, ground infrastructure, satellite handovers, terminal power, and weather-sensitive link availability [25–27]. The architecture therefore treats LEO as an optional resilience path, not as a core dependency. LEO selection cannot bypass identity, policy, isolation, or accounting.

## 2.5 Graph Learning and Wireless Resource Allocation

Shared Wi-Fi access is naturally graph-structured. APs interfere, subscribers associate, channels overlap, and mobility changes edge weights over time. GCN and GAT architectures provide learning mechanisms for graph-structured data [14, 15]. GNN-based radio resource management has direct precedent in wireless systems research [16]. This proposal, however, restricts graph learning to slow macro-policy optimisation and never allows it to override fast local safety enforcement.

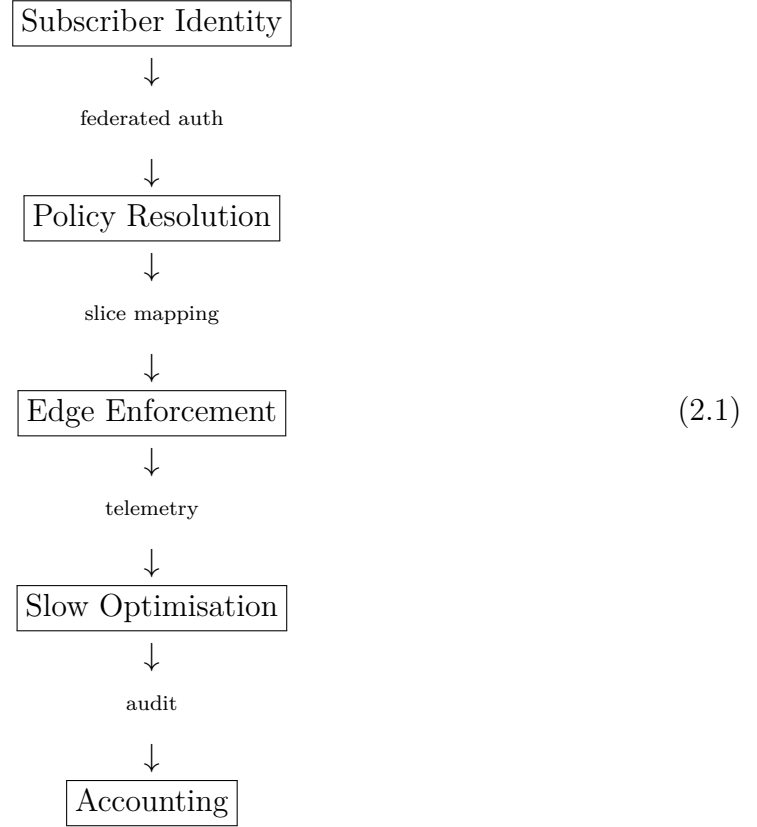
## 2.6 Theoretical Framework

The theoretical framework integrates four traditions: network access control, queueing and effective-bandwidth theory, graph learning, and safe constrained control. The revised proposal avoids treating queueing approximations, large-deviation bounds, and neural-network optimisation as interchangeable. Each is assigned a specific role:

1. AAA and 802.1X define subscriber identity and policy authority.
2. Finite-queue models describe implementation behaviour; auxiliary infinite-buffer models support large-deviation overflow estimation.

3. GNNs update slow macro policies from graph telemetry.
4. State-dependent CMDP safe action sets ensure that learning-based decisions never bypass authentication, policy, isolation, accounting, or resource feasibility.

The system chain is:



# Chapter 3

## System Architecture

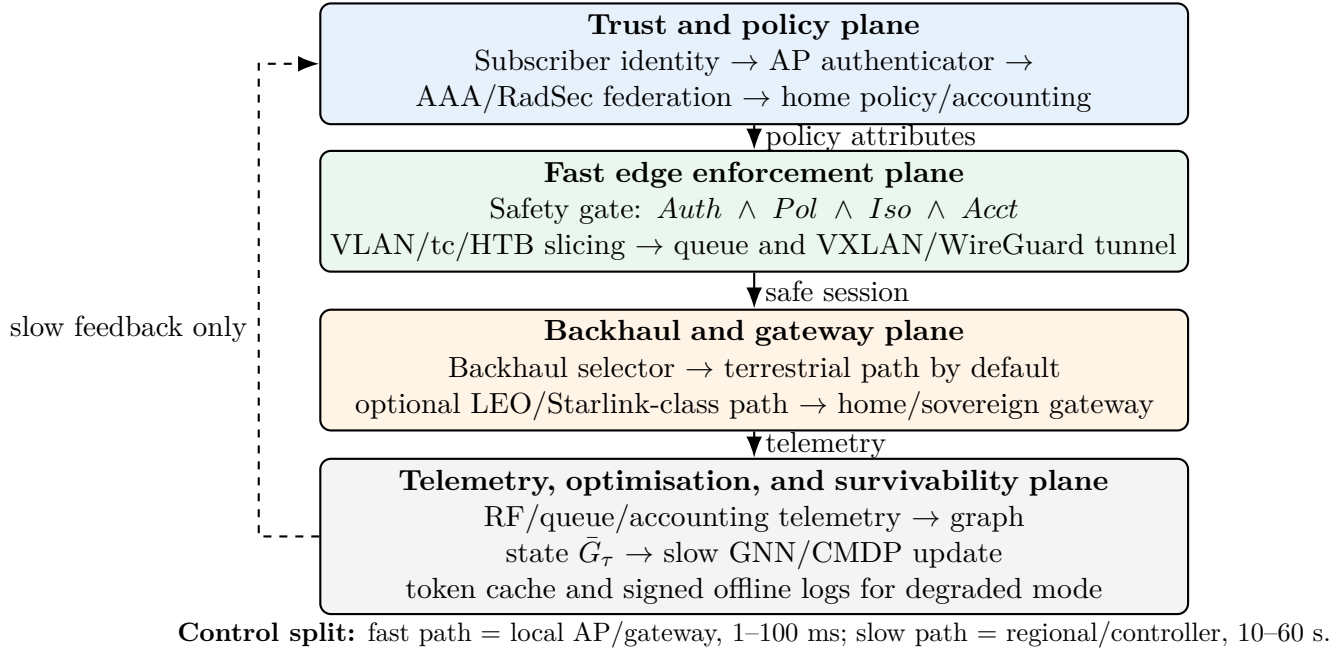
### 3.1 Architectural Boundary Conditions

The architecture is bounded by six deployment constraints:

1. Every participating AP must be administratively enrolled.
2. Subscriber identity must be stronger than a shared Wi-Fi password; certificate-based or profile-based EAP is the minimum proposed mechanism.
3. The home domain remains the authoritative source of subscriber policy.
4. Authentication, policy compliance, slice isolation, accounting consistency, and edge-resource feasibility are hard feasibility requirements.
5. Commodity edge APs are treated as constrained compute platforms.
6. Machine learning operates only in the slow control path and cannot override local safety rules.

### 3.2 Clean Four-Plane Architecture

The architecture is organised into four planes: trust and policy, fast edge enforcement, backhaul and gateway selection, and telemetry/optimisation/survivability. Figure 3.1 presents the simplified structure used throughout the rest of the proposal.



**Figure 3.1:** Clean layered subscriber-centric broadband architecture. The packet fast path remains local at the edge AP, while the GNN/CMDP block only issues slow macro-policy updates. Optional LEO backhaul is isolated to the backhaul plane and cannot bypass the safety gate.

**Trust and policy plane.** Handles federated AAA, RADSEC transport, home-policy lookup, access decisions, and accounting identity initialisation.

**Fast edge enforcement plane.** Executes packet forwarding, VLAN/VXLAN separation, Linux tc/HTB/fq\_codel shaping, local overload protection, and isolation checks.

**Backhaul and gateway plane.** Selects terrestrial or optional LEO backhaul paths after authentication and slice assignment have succeeded.

**Telemetry, optimisation, and survivability plane.** Aggregates telemetry into graph states, performs slow macro-policy optimisation, and supports token-cache degraded mode during local isolation.

### 3.3 Fast Path, Slow Path, and Control Epochs

The two-timescale model prevents the GNN controller from chasing millisecond-scale packet bursts. The local AP and edge gateway enforce existing rules at fast timescales, while the GNN updates macro policies only after telemetry aggregation.

$$T_{\text{fast}} \ll T_{\text{GNN}}, \quad t_{\text{infer}} \leq 150 \text{ ms} \ll T_{\text{GNN}}. \quad (3.1)$$



**Table 3.1:** Control-plane split and temporal responsibilities.

| Layer           | Location            | Timescale  | Responsibility                                                                     |
|-----------------|---------------------|------------|------------------------------------------------------------------------------------|
| Fast path       | Edge AP/gateway     | 1–100 ms   | Enforce slice rules, queue shaping, VLAN/tunnel isolation, overload protection.    |
| Local guard     | Edge AP/gateway     | 100 ms–1 s | Apply deterministic fallback under overload, queue instability, or backhaul loss.  |
| GNN control     | Regional controller | 10–60 s    | Update association preferences, channel plans, slice weights, and backhaul policy. |
| Offline trainer | Research server     | Hours–days | Train and validate graph models from telemetry and simulation traces.              |

### 3.4 Commodity Edge Resource Budget

Each edge AP is modelled as a constrained compute platform with resource vector

$$\mathbf{R}_j^{\text{edge}} = (CPU_j, RAM_j, IRQ_j, Crypto_j, Encaps_j, Queue_j, Tel_j). \quad (3.2)$$

The feasibility constraint is

$$CPU_j \leq CPU_j^{\max}, \quad RAM_j \leq RAM_j^{\max}, \quad IRQ_j \leq IRQ_j^{\max}, \quad \forall j. \quad (3.3)$$

The operational acceptance target is  $CPU_j \leq 0.75CPU_j^{\max}$  under peak traffic, leaving headroom for control-plane tasks. This target is an empirical acceptance criterion, not a theoretical guarantee.

#### 3.4.1 Driver Buffer Admission Heuristic

The driver-ring control is an empirical admission-control heuristic, not a stability theorem. Let  $\Lambda_{k,j}$  be the measured normalised frame load of slice  $k$  on AP  $a_j$ . The admission condition is

$$\Omega_j = \sum_{k \in \mathcal{K}} \Lambda_{k,j} \leq \Omega_j^{\max} \left[ 1 - \xi \frac{Drop_{driver,j}}{Arrival_j + \epsilon} \right]_+, \quad (3.4)$$

where  $[x]_+ = \max(0, x)$ ,  $\epsilon > 0$  prevents division by zero, and  $\xi$  is selected through hardware calibration. This prevents the earlier draft from asserting an unproven universal ring-buffer stability condition.

### 3.5 Edge Survivability Under Local Isolation

Nepal-specific power cuts, fibre disruptions, and intermittent backhaul motivate a three-state survivability model:

$$\omega_j \in \{\text{Online}, \text{Degraded}, \text{Isolated}\}. \quad (3.5)$$

In *Online* mode, all authentication and accounting flows use the normal AAA/RadSec path. In *Degraded* mode, previously authorised subscribers may use short-lived cached tokens with strict bandwidth caps and signed offline accounting. In *Isolated* mode, no privilege escalation is possible; the AP provides either no subscriber service or limited guest-degraded service according to local policy.

A cached token is

$$\tau_i = \text{HMAC-SHA256}_{K_{\text{net}}}(ID_i \| TS_i \| SLA_i), \quad (3.6)$$

with anti-replay timestamp  $TS_i$ , service entitlement bitmap  $SLA_i$ , and key  $K_{\text{net}}$  provisioned before failure. The CPU target for re-authentication storms is treated as an empirical acceptance criterion:

$$CPU_{\text{auth},j} \leq 0.05 CPU_j^{\text{max}} \quad \text{during a 100-client cached-token re-authentication test.} \quad (3.7)$$

### 3.6 Normalised LEO Control Backoff

The clipping operator is defined as

$$\text{clip}_{[a,b]}(x) = \min\{b, \max\{a, x\}\}. \quad (3.8)$$

The LEO-specific macro-epoch backoff uses dimensionless inputs:

$$\Delta t_{\text{eff}}^{\text{LEO}} = \text{clip}_{[\Delta t_{\text{macro}}, \Delta t_{\text{max}}]} \left( \Delta t_{\text{macro}} \exp \left[ \kappa_{\text{RTT}} \frac{\sigma_{\text{RTT}}^2}{\sigma_{\text{RTT,ref}}^2 + \epsilon} + \kappa_{\text{drop}} \frac{\text{Drop}_{\text{LEO}}}{\text{Drop}_{\text{ref}} + \epsilon} \right] \right). \quad (3.9)$$

The clipping operation prevents exponential blow-up, while normalisation prevents dimensional inconsistency inside the exponential. The coefficients  $\kappa_{\text{RTT}} \geq 0$  and  $\kappa_{\text{drop}} \geq 0$  are LEO-backoff sensitivity parameters. They are not LEO cost weights and are therefore not constrained to sum to one.

### 3.7 Systems Verification Matrix

**Table 3.2:** Systems verification and boundary performance matrix.

| Element                | Environment               | Acceptance Target                                                         |
|------------------------|---------------------------|---------------------------------------------------------------------------|
| Slice isolation        | Emulation + prototype     | Zero cross-slice packets matching foreign VLAN/VNI tags.                  |
| Edge compute ceiling   | Hardware benchmark        | $CPU_j \leq 0.75 CPU_j^{\max}$ under peak configured workload.            |
| Auth-storm resistance  | Hardware benchmark        | $CPU_{auth,j} \leq 0.05 CPU_j^{\max}$ as empirical target, not guarantee. |
| GNN inference          | Trace simulation          | $t_{infer} \leq 150 \text{ ms}$ and $t_{infer} \ll T_{\text{GNN}}$ .      |
| Wi-Fi path-loss model  | RSSI survey               | RMSE target 4–6 dB against floor-plan/RSSI heatmap data.                  |
| LEO model              | Backhaul tests/simulation | LEO results reported separately from terrestrial-only baseline.           |
| Accounting consistency | Prototype + emulation     | $ Bytes_{RADIUS} - Bytes_{GW}  \leq \varepsilon_{acct}$ .                 |

# Chapter 4

## Formal Mathematical Model

### 4.1 System Variables

Let  $\mathcal{S} = \{s_1, \dots, s_N\}$  denote subscribers and  $\mathcal{A} = \{a_1, \dots, a_M\}$  participating access points. Let  $\mathcal{K}$  denote traffic slices,  $\mathcal{D}$  administrative domains, and  $\mathcal{H}$  available backhaul modes. Subscriber  $s_i$  is represented by

$$s_i = (id_i, d_i, p_i, b_i^{min}, b_i^{max}, \ell_i^{max}, \delta_i^{max}, \alpha_i, \rho_i), \quad (4.1)$$

where  $id_i$  is pseudonymised identity,  $d_i$  home domain,  $p_i$  policy profile,  $b_i^{min}, b_i^{max}$  bandwidth bounds,  $\ell_i^{max}$  latency bound,  $\delta_i^{max}$  loss bound,  $\alpha_i$  service class, and  $\rho_i$  accounting identifier.

AP  $a_j$  is represented by

$$a_j = (cap_j, bh_j, c_j, \eta_j, \phi_j, \omega_j, \mathbf{R}_j^{\text{edge}}), \quad (4.2)$$

where  $cap_j$  is wireless capacity,  $bh_j$  effective backhaul capacity,  $c_j$  channel,  $\eta_j$  utilisation,  $\phi_j$  local policy,  $\omega_j$  survivability state, and  $\mathbf{R}_j^{\text{edge}}$  edge resource vector.

The assignment variables are

$$x_{ij} = \begin{cases} 1, & s_i \text{ is served by } a_j, \\ 0, & \text{otherwise,} \end{cases} \quad z_{ik} = \begin{cases} 1, & s_i \text{ is mapped to slice } k, \\ 0, & \text{otherwise.} \end{cases} \quad (4.3)$$

A subscriber session is valid only if

$$Valid(o_i) = 1 \iff Auth_i = 1 \wedge Pol_i = 1 \wedge Iso_i = 1 \wedge Acct_i = 1. \quad (4.4)$$

### 4.1.1 Notation, Parameter Selection, and Normalisation Convention

To avoid ambiguity between modelling variables, calibrated coefficients, and experimentally fixed thresholds, this section defines the notation and parameter-selection conventions used throughout the mathematical model and evaluation framework. Parameters are classified into four groups: operational policy parameters, measured quantities, calibrated coefficients, and pre-registered evaluation thresholds.

**Table 4.1:** Additional notation and parameter-selection conventions.

| Symbol                 | Meaning                                                                       | Selection method                                       |
|------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------|
| $T_{\text{rot}}$       | Nominal rotation period of the cached-token signing key $K_{\text{net}}$ .    | Operational security policy.                           |
| $T_{\text{token}}$     | Maximum validity window of a cached offline-access token.                     | Chosen such that $T_{\text{token}} < T_{\text{rot}}$ . |
| $\Delta_{\text{min}}$  | Minimum practically meaningful improvement in the composite evaluation score. | Fixed before comparative experiments.                  |
| $B_{\text{curr}}$      | Current measured available backhaul capacity during a telemetry epoch.        | Measured from AP/gateway telemetry.                    |
| $B_{\text{nominal}}$   | Nominal expected backhaul capacity for the AP or scenario class.              | Configured baseline or pilot-free engineering value.   |
| $\kappa_B$             | Sensitivity of telemetry interval to backhaul degradation.                    | Calibrated coefficient.                                |
| $\kappa_{\text{CPU}}$  | Sensitivity of telemetry interval to CPU pressure.                            | Calibrated coefficient.                                |
| $\kappa_{\text{RTT}}$  | Sensitivity of the LEO macro-epoch backoff to RTT variance.                   | Calibrated coefficient.                                |
| $\kappa_{\text{drop}}$ | Sensitivity of the LEO macro-epoch backoff to packet-drop rate.               | Calibrated coefficient.                                |

The cached-token key rotation parameter  $T_{\text{rot}}$  is treated as an operational security-policy parameter rather than a queueing or optimisation parameter. Cached-token validity is bounded by a shorter token lifetime  $T_{\text{token}}$ , with

$$T_{\text{token}} < T_{\text{rot}}. \quad (4.5)$$

During local degraded operation, cached-token authentication is rejected when the locally held key age exceeds  $2T_{\text{rot}}$ . This conservative bound prevents prolonged use of stale token-signing material while allowing short temporary service continuity during backhaul or AAA unavailability.

The threshold  $\Delta_{\text{min}}$  in the acceptance criterion is fixed before comparative baseline

experiments are performed. It represents the minimum practically meaningful improvement in the normalised composite score  $S(M)$ , not a value tuned after observing results. Unless otherwise justified by pilot-free engineering constraints, the default value is

$$\Delta_{\min} = 0.05. \quad (4.6)$$

Sensitivity analysis will report whether the main conclusions change for

$$\Delta_{\min} \in \{0.02, 0.05, 0.10\}. \quad (4.7)$$

All convex weighted sums use non-negative weights normalised to one. Specifically, the graph-interference weights  $\beta_r$ , CMDP cost weights  $\omega_r$ , LEO cost weights  $\psi_r$ , and composite-score weights  $\zeta_r$  satisfy

$$\beta_r \geq 0, \quad \sum_r \beta_r = 1, \quad (4.8)$$

$$\omega_r \geq 0, \quad \sum_r \omega_r = 1, \quad (4.9)$$

$$\psi_r \geq 0, \quad \sum_r \psi_r = 1, \quad (4.10)$$

$$\zeta_r \geq 0, \quad \sum_r \zeta_r = 1. \quad (4.11)$$

These weights are fixed within a declared scenario class before comparative evaluation. They are not claimed to be globally transferable across dense urban, semi-urban, rural-extension, and mountain-backhaul-constrained scenarios without recalibration.

The telemetry-throttle and LEO-backoff coefficients are treated differently from convex cost weights. The coefficients  $\kappa_B$ ,  $\kappa_{\text{CPU}}$ ,  $\kappa_{\text{RTT}}$ , and  $\kappa_{\text{drop}}$  are non-negative sensitivity parameters:

$$\kappa_B \geq 0, \quad \kappa_{\text{CPU}} \geq 0, \quad \kappa_{\text{RTT}} \geq 0, \quad \kappa_{\text{drop}} \geq 0. \quad (4.12)$$

They control how strongly telemetry sampling and LEO macro-epoch backoff react to backhaul degradation, CPU pressure, RTT variance, and packet-drop rate, respectively. They are calibrated through controlled sensitivity analysis and are not required to sum to one.

## 4.2 Finite Implementation Queue and Auxiliary LDT Queue

The model separates the physical finite-buffer implementation queue from the auxiliary infinite-buffer queue used for large-deviation analysis. This avoids the contradiction of

imposing a finite upper boundary  $B_k$  while later evaluating  $P(Q_k > B_k)$ .

#### 4.2.1 Finite Drop-Tail Implementation Model

For slice  $k$ , the implemented queue is a reflected/censored workload process  $Q_k(t) \in [0, B_k]$ . The service rate used in the queueing model is not the nominal configured traffic-control rate alone. Instead, the effective bottleneck service capacity is

$$C_k^{eff}(t) = \min\{C_k^{tc}(t), C_k^{air}(t), C_k^{cpu}(t), C_k^{tun}(t), C_k^{bh}(t)\}, \quad (4.13)$$

where  $C_k^{tc}(t)$  is the configured traffic-control service rate,  $C_k^{air}(t)$  is the airtime-limited wireless capacity,  $C_k^{cpu}(t)$  is the CPU-limited forwarding capacity,  $C_k^{tun}(t)$  is the tunnel-limited capacity after encapsulation or encryption overhead, and  $C_k^{bh}(t)$  is the available backhaul capacity.

The finite-buffer workload process is therefore modelled as

$$dQ_k(t) = (\lambda_k(t) - C_k^{eff}(t)) dt + \sigma_k dW_t + dR_k^0(t) - dL_k^B(t), \quad (4.14)$$

where  $R_k^0(t)$  is the lower-bound reflection process at zero and  $L_k^B(t)$  is the upper-bound loss/local-time process at buffer threshold  $B_k$ . Overflow in the implementation model is measured through the loss process  $L_k^B(t)$ , not through the impossible event  $Q_k(t) > B_k$ .

The associated probability-flux expression for the unconstrained interior is

$$J(q, t) = \mu_k(t)p(q, t) - \frac{\sigma_k^2}{2} \frac{\partial p(q, t)}{\partial q}, \quad \mu_k(t) = \lambda_k(t) - C_k^{eff}(t). \quad (4.15)$$

At  $q = 0$ , reflection imposes  $J(0, t) = 0$ . At  $q = B_k$ , finite-buffer packet loss is recorded through the loss process rather than by claiming that the physical queue exceeds  $B_k$ . All queueing, effective-bandwidth, and SLA calculations use  $C_k^{eff}(t)$ , not the nominal configured traffic-control rate.

#### 4.2.2 Diffusion Coefficient Estimation

The diffusion coefficient must be a variance rate, not a raw telemetry-window variance. Let  $A_k(t)$  denote cumulative input to slice  $k$ . For macro epoch  $\tau$ , define the quasi-static interval

$$T_\tau = [\tau T_{GNN}, (\tau + 1)T_{GNN}). \quad (4.16)$$

Within  $T_\tau$ , the empirical diffusion estimate is

$$\hat{\sigma}_{k,\tau}^2 = \frac{1}{\Delta} \text{Var} [A_k(t + \Delta) - A_k(t) - \hat{\lambda}_{k,\tau}\Delta], \quad t, t + \Delta \in T_\tau, \quad (4.17)$$

where  $\Delta$  is the measurement interval and  $\hat{\lambda}_{k,\tau}$  is the estimated mean input rate for slice  $k$  during epoch  $\tau$ . This definition gives units of buffer-units squared per second and is compatible with the drift-diffusion approximation.

The diffusion coefficient  $\sigma_k$  in Eq. (4.14) is interpreted at the macro-epoch scale. In implementation and empirical estimation,  $\sigma_k^2$  is instantiated by the epoch-indexed estimator  $\hat{\sigma}_{k,\tau}^2$  in Eq. (4.17) under the quasi-static assumption that arrival and effective-service statistics remain approximately stationary within  $T_\tau$ .

### 4.3 Large Deviations Bound for the Auxiliary Infinite Queue

Large Deviations Theory is applied to an auxiliary infinite-buffer workload  $Q_k^\infty(t)$ , not to the finite implementation queue. The virtual tail probability

$$P(Q_k^\infty > B_k) \quad (4.18)$$

approximates the overflow risk that the finite queue would experience at threshold  $B_k$ .

Let  $A_k(t)$  be the cumulative input process. Its effective bandwidth is

$$E_k(\theta) = \lim_{t \rightarrow \infty} \frac{1}{\theta t} \log \mathbb{E} \left[ e^{\theta A_k(t)} \right], \quad \theta > 0. \quad (4.19)$$

The stability condition is  $E_k(0^+) < C_k^{eff}$ , where  $E_k(0^+)$  is the mean input rate when the limit exists. The decay exponent is defined safely as

$$\theta_k^* = \sup\{\theta > 0 : E_k(\theta) < C_k^{eff}\}. \quad (4.20)$$

If  $E_k(\theta)$  is continuous, strictly increasing, and crosses  $C_k^{eff}$ , then the supremum is the unique positive solution of

$$E_k(\theta_k^*) = C_k^{eff}. \quad (4.21)$$

Otherwise, the analysis reports the supremum-based bound rather than assuming root existence.

Under the effective-bandwidth assumptions, the auxiliary workload satisfies the logarithmic asymptotic relation

$$\lim_{B_k \rightarrow \infty} \frac{1}{B_k} \log P(Q_k^\infty > B_k) = -\theta_k^*. \quad (4.22)$$

For finite  $B_k$ , the quantity  $e^{-\theta_k^* B_k}$  is used only as an exponential-order overflow-risk indicator, not as a calibrated finite-buffer loss probability unless validated against measured queue-loss traces.



This bound estimates overflow risk only. Cross-slice isolation is not a consequence of large-deviation theory; it is enforced by VLAN/VNI separation, queue-discipline configuration, and verified packet tracing in the edge datapath.

## 4.4 Hard Safety Feasibility and State-Dependent Safe Actions

**Definition 4.1** (State-dependent safe action set). For a state  $x \in \mathcal{X}$ , the safe action set is

$$\mathcal{U}_{\text{safe}}(x) = \{u \in \mathcal{U}(x) : \text{Auth}(x, u) = 1, \text{Pol}(x, u) = 1, \text{Iso}(x, u) = 1, \text{Acct}(x, u) = 1, \text{EdgeFeas}(x, u) = 1\} \quad (4.23)$$

The edge-feasibility predicate is

$$\begin{aligned} \text{EdgeFeas}(x, u) = 1 \iff & \text{CPU}_j \leq \text{CPU}_j^{\max} \wedge \text{RAM}_j \leq \text{RAM}_j^{\max} \wedge \text{IRQ}_j \leq \text{IRQ}_j^{\max} \\ & \wedge \Omega_j \leq \Omega_j^{\max} \wedge C_k^{\text{eff}} > E_k(0^+). \end{aligned} \quad (4.24)$$

The final condition ensures that the effective service capacity of the selected slice exceeds the mean input rate implied by the effective-bandwidth model. Thus, a safe action is not merely authenticated and policy-valid; it must also be feasible under measured edge-resource and queue-stability constraints.

**Definition 4.2** (Safe policy set). A policy is safe if it chooses only safe actions in every reachable state:

$$\Pi_{\text{safe}} = \{\pi : \pi(x) \in \mathcal{U}_{\text{safe}}(x), \forall x \in \mathcal{X}\}. \quad (4.25)$$

Basic access constraints are

$$\sum_{j=1}^M x_{ij} \leq 1, \quad \forall i, \quad (4.26)$$

$$x_{ij} \leq \text{Auth}_i, \quad \forall i, j, \quad (4.27)$$

$$z_{ik} \leq \text{Pol}_{ik}, \quad \forall i, k. \quad (4.28)$$

Capacity and QoS constraints are

$$\sum_{i=1}^N x_{ij} b_i \leq \min(\text{cap}_j, b h_j), \quad \forall j, \quad (4.29)$$

$$b_i^{\min} \leq b_i \leq b_i^{\max}, \quad \ell_i \leq \ell_i^{\max}, \quad \delta_i \leq \delta_i^{\max}. \quad (4.30)$$

Edge feasibility follows Eq. (4.24).

## 4.5 Interference Graph and Edge-Aware Attention

The graph state is

$$G_t = (V_t, E_t, X_t, E_t^{feat}), \quad (4.31)$$

where  $X_t$  stores node features and  $E_t^{feat}$  stores edge features, including interference weight, distance proxy, channel overlap, and utilisation coupling. The AP-AP interference edge weight is

$$w_{jl}(t) = \beta_1 \mathbb{1}[c_j = c_l] + \beta_2 \frac{1}{(d_{jl} + \epsilon_d)^{\hat{\gamma}}} + \beta_3(\eta_j + \eta_l) + \beta_4 O_{jl}, \quad (4.32)$$

with normalised terms,  $\beta_r \geq 0$ , and  $\sum_r \beta_r = 1$ .

Each scalar edge feature is min-max normalised within the scenario class before training:

$$\bar{x}_{r,jl} = \frac{x_{r,jl} - \min_{(u,v) \in E} x_{r,uv}}{\max_{(u,v) \in E} x_{r,uv} - \min_{(u,v) \in E} x_{r,uv} + \epsilon}. \quad (4.33)$$

The calibrated or learned coefficients  $\beta_r$  are therefore comparable only within a declared scenario class. They are not claimed to be globally transferable without re-normalisation or recalibration.

The attention mechanism is edge-aware:

$$\alpha_{uv}^{(l)} = \text{softmax}_u \left( \text{LeakyReLU} \left( \mathbf{a}^{(l)\top} \left[ W^{(l)} h_v^{(l)} \| W^{(l)} h_u^{(l)} \| e_{uv}^{(l)} \right] \right) \right), \quad (4.34)$$

where  $e_{uv}^{(l)}$  is the edge-feature vector. This corrects the earlier ambiguity of hiding interference weights inside node features.

## 4.6 Physics-Informed Path-Loss Scope

Indoor Wi-Fi is modelled with a log-distance multi-wall model:

$$PL(d) = PL_0 + 10\hat{\gamma} \log_{10} \left( \frac{d}{d_0} \right) + \sum_{m=1}^{M_w} N_{w,m} L_{w,m}. \quad (4.35)$$

This model is validated against floor-plan/RSSI heatmap data, not terrain grid data. The empirical acceptance target is  $RMSSE_{RSSI} \leq 4\text{--}6$  dB unless stronger measurement evidence justifies a tighter bound.

For LEO or long-range rural backhaul, the physics-informed loss may use ITU-R P.838/P.618 rain and earth-space propagation constraints [23, 24]:

$$\mathcal{L}_{PINN} = \mathcal{L}_{data} + \lambda_{phys} \mathcal{L}_{phys}, \quad \mathcal{L}_{phys} = \frac{1}{Q} \sum_{q=1}^Q (\hat{y}_q - y_{phys,q})^2. \quad (4.36)$$

Applying ITU-R rain attenuation to ordinary indoor Wi-Fi would be outside this proposal's valid scope.

## 4.7 Consistent Cost-Minimisation CMDP

The slow controller is formulated as a cost-minimising CMDP:

$$\mathcal{M} = (\mathcal{X}, \mathcal{U}_{\text{safe}}(\cdot), P, C, \gamma), \quad \gamma \in (0, 1). \quad (4.37)$$

The state  $x$  is a high-dimensional graph-embedding and telemetry summary, so the Bellman equation is written with an expectation rather than a discrete summation:

$$V^*(x) = \min_{u \in \mathcal{U}_{\text{safe}}(x)} \left[ C(x, u) + \gamma \mathbb{E}_{x' \sim P(\cdot|x, u)} V^*(x') \right]. \quad (4.38)$$

If  $\mathcal{U}_{\text{safe}}(x) = \emptyset$ , the minimisation in Eq. (4.38) is not evaluated over an empty action set. Instead, the controller invokes the deterministic survivability fallback associated with the AP state  $\omega_j$ . In the Online state, new admissions are rejected while existing valid sessions are preserved only if they remain feasible. In the Degraded state, only previously authorised subscribers with valid cached tokens may receive capped service and signed offline accounting. In the Isolated state, no subscriber privilege escalation is permitted; the AP either denies subscriber service or applies an explicitly configured guest-degraded local policy.

The instantaneous cost is defined as

$$C(x, u) = \omega_1 \mathcal{I}(G) + \omega_2 V_{SLA} + \omega_3 O_{tun} + \omega_4 O_{edge} + \omega_5 O_{LEO} - \omega_6 T_{norm} - \omega_7 J(\tilde{\mathbf{b}}). \quad (4.39)$$

Here  $T_{norm}$  is normalised delivered throughput. Latency, loss, and minimum-rate violations are not separately embedded inside  $T_{norm}$  because they are already represented through  $V_{SLA}$ . This prevents the cost function from double-counting the same service degradation through both utility and SLA-violation terms.

**Table 4.2:** Operational definitions of CMDP cost terms.

| Term             | Operational definition                                                                      | Evidence source                         |
|------------------|---------------------------------------------------------------------------------------------|-----------------------------------------|
| $\mathcal{I}(G)$ | Normalised graph-interference cost computed from active AP–AP interference edge weights.    | Radio-topology matrix.                  |
| $V_{SLA}$        | Fraction of subscriber-epochs violating latency, loss, or minimum-rate SLA bounds.          | Slice-performance matrix.               |
| $O_{tun}$        | Tunnel overhead ratio $(Bytes_{tunnel} - Bytes_{payload}) / (Bytes_{payload} + \epsilon)$ . | Slice-performance and gateway counters. |
| $O_{edge}$       | Normalised edge-resource pressure over CPU, RAM, IRQ, crypto, queueing, and telemetry load. | Edge-resource matrix.                   |
| $O_{LEO}$        | Normalised backhaul penalty for LEO path RTT, jitter, outage, handover, and power cost.     | Backhaul-resilience matrix.             |
| $T_{norm}$       | Delivered throughput normalised by subscriber entitlement or configured maximum.            | Slice-performance matrix.               |

If a PPO-style implementation requires a reward signal, it is defined only as

$$R(x, u) = -C(x, u). \quad (4.40)$$

This removes the earlier reward/cost sign-convention mismatch.

Because the state is represented by continuous telemetry and graph embeddings, any PPO-style implementation uses function approximation. Therefore, tabular CMDP convergence guarantees do not directly apply. Safety is enforced by the state-dependent safe action filter rather than learned solely from reward penalties.

Fairness is measured using Jain’s index over normalised subscriber entitlement:

$$J(\tilde{\mathbf{b}}) = \frac{(\sum_{i=1}^N \tilde{b}_i)^2}{N \sum_{i=1}^N \tilde{b}_i^2}, \quad \tilde{b}_i = \frac{b_i}{b_i^{max}}. \quad (4.41)$$

## 4.8 LEO Backhaul Resilience Cost

For backhaul mode  $h$ , effective capacity is

$$bh_{j,t}^{eff} = A_{j,t}^h C_{j,t}^h, \quad (4.42)$$

where  $A_{j,t}^h \in \{0, 1\}$  is path availability and  $C_{j,t}^h$  is usable capacity.

For a LEO path, the cost uses normalised dimensionless terms:

$$C_{LEO,j,t}^{cost} = \psi_1 \frac{RTT_{j,t}}{RTT_{ref}} + \psi_2 \frac{Jit_{j,t}}{Jit_{ref}} + \psi_3 P_{out,j,t} + \psi_4 Handover_{j,t} + \psi_5 EnergyCost_{j,t}. \quad (4.43)$$

The energy-cost term is defined as

$$EnergyCost_{j,t} = 1 - \frac{PwrAvail_{j,t}}{PwrMax_j + \epsilon}. \quad (4.44)$$

Thus, lower available terminal power increases the LEO cost. This prevents the backhaul selector from accidentally treating high battery or power availability as a penalty.

The selector chooses

$$h_{j,t}^* = \arg \min_{h \in \mathcal{H}} C_{h,j,t}^{cost} \quad \text{subject to } \pi \in \Pi_{\text{safe}}. \quad (4.45)$$

# Chapter 5

## Methodology and Evaluation Framework

### 5.1 Research Design

The study uses design-science systems research [30], controlled experimentation, emulation, simulation, and contextual validation. The artifact is the safety-constrained subscriber-centric broadband architecture. Evaluation tests whether the artifact satisfies authentication, isolation, QoS, accounting, edge feasibility, optimisation, LEO-assisted resilience, and Nepal-context deployment requirements.

Five evidence tracks are defined:

**Track 1 - Prototype:** OpenWrt + FreeRADIUS + VLAN/`tc` + WireGuard/VXLAN.

Provides measured ground truth for authentication latency, slicing overhead, and AP resource consumption.

**Track 2 - Edge overhead:** Hardware-in-the-loop benchmarking on constrained APs under slice saturation.

**Track 3 - Emulation:** Mininet-WiFi topologies calibrated against prototype measurements.

**Track 4 - Simulation and optimisation:** NS-3 and PyTorch Geometric scenarios evaluating graph-based macro policy.

**Track 5 - Contextual validation:** Nepal scenario mapping, NTA regulatory compatibility, and deployment plausibility.

## 5.2 Dataset Design

All records are classified as Measured, Emulated, Simulated, or Contextual (M/E/S/C). Claims cannot be extended beyond their evidence class.

**Table 5.1:** Dataset matrices, variables, and analytical role.

| Matrix                              | Core Variables                                                                                                                              | Analytical Role                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Identity-session matrix (M/E)       | Subscriber pseudonym, realm, visited AP, home domain, EAP method, auth result, failure code, auth latency, session duration, accounting ID. | Tests whether identity operates as primary access-control primitive.    |
| Slice-performance matrix (M/E)      | Slice ID, VLAN/VNI, rate cap, throughput, latency, jitter, loss, queue occupancy, tunnel overhead, SLA status.                              | Tests subscriber-specific QoS and slice enforcement.                    |
| Edge-resource matrix (M)            | CPU, RAM, IRQ rate, crypto throughput, encapsulation overhead, queueing cost, telemetry cost.                                               | Determines AP feasibility under resource constraints.                   |
| Radio-topology matrix (M/E/S)       | AP ID, subscriber ID, RSSI, SNR, channel, retry rate, channel utilisation, edge weight, wall class.                                         | Constructs graph states and path-loss validation data.                  |
| Queueing matrix (M/E/S)             | Arrival increments $A_k(t + \Delta) - A_k(t)$ , service rate $C_k$ , buffer threshold $B_k$ , loss process increments $L_k^B$ .             | Estimates diffusion coefficient, overflow risk, and finite-buffer loss. |
| Backhaul-resilience matrix (M/E)    | Backhaul mode, RTT, jitter, packet loss, outage, LEO handover risk, terminal power, failover time.                                          | Evaluates terrestrial and LEO resilience.                               |
| Accounting-consistency matrix (M/E) | RADIUS counters, AP counters, gateway counters, byte mismatch, termination cause, offline signature.                                        | Tests normal and degraded-mode auditability.                            |

| Matrix                           | Core Variables                                                                                          | Analytical Role                                                      |
|----------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Control-loop matrix (M/E/S)      | GNN inference time, telemetry aggregation delay, policy update latency, epoch staleness, rollback rate. | Validates two-timescale separation.                                  |
| Contextual deployment matrix (C) | Location class, terrain class, backhaul type, power reliability proxy, regulatory constraint.           | Maps Nepal-specific conditions to emerging-economy scenario classes. |

### 5.3 Data-Plane Tooling Hierarchy

Commodity baseline enforcement uses Linux `tc`, `HTB`, `fq_codel`, VLANs, and optionally VXLAN/WireGuard depending on the benchmark path. eBPF/XDP is not assumed on all APs; it is evaluated only on premium ARMv8/x86 nodes as an advanced benchmark track. This prevents the proposal from claiming low-end feasibility for mechanisms that may require stronger kernel, driver, and CPU support.

### 5.4 System-Level Telemetry Throttle

To prevent telemetry from competing with user-plane data, the system-level telemetry interval is

$$\Delta t_{\text{eff}}^{\text{sys}} = \text{clip}_{[\Delta t_{\text{base}}, \Delta t_{\text{sys, max}}]} \left( \Delta t_{\text{base}} \exp \left[ \kappa_B \left[ 1 - \frac{B_{\text{curr}}}{B_{\text{nominal}} + \epsilon} \right]_+ + \kappa_{\text{CPU}} \frac{\text{CPU}_j(t)}{\text{CPU}_j^{\text{max}}} \right] \right), \quad (5.1)$$

where  $[x]_+ = \max(0, x)$ . The coefficient  $\kappa_B$  controls sensitivity to backhaul degradation, while  $\kappa_{\text{CPU}}$  controls sensitivity to AP CPU pressure. These are calibrated non-negative sensitivity coefficients and are not required to sum to one. The positive-part operator prevents the telemetry controller from increasing sampling frequency merely because transient throughput exceeds the nominal baseline. This equation is separate from the LEO-specific backoff in Eq. (3.9).

### 5.5 Experimental Baselines

1. B1 - WPA-PSK router-centric baseline.
2. B2 - Captive portal baseline.



3. B3 - VLAN-only static policy baseline.
4. B4 - Greedy AP/channel allocation baseline.
5. B5 - DQN-style unconstrained learning baseline.
6. B6 - Vanilla GNN without state-dependent safety filtering.
7. B7 - Terrestrial-only backhaul and LEO-assisted backhaul variants reported separately.

A configuration is accepted only if it improves the composite performance score while satisfying all hard feasibility constraints:

$$Accept(\mathcal{M}) = 1 \iff S(\mathcal{M}) - S(\mathcal{B}) > \Delta_{\min} \wedge \pi \in \Pi_{\text{safe}} \wedge \mathbf{R}_j^{\text{edge}} \leq \mathbf{R}_j^{\max}. \quad (5.2)$$

The composite score is

$$S(\mathcal{M}) = \zeta_1 T_{5p}(\mathcal{M}) - \zeta_2 L_{95p}(\mathcal{M}) - \zeta_3 V_{\text{SLA}}(\mathcal{M}) + \zeta_4 J(\tilde{\mathbf{b}}) - \zeta_5 O_{\text{edge}}(\mathcal{M}), \quad (5.3)$$

where  $T_{5p}$  is fifth-percentile throughput,  $L_{95p}$  is ninety-fifth-percentile latency,  $V_{\text{SLA}}$  is the SLA violation rate,  $J(\tilde{\mathbf{b}})$  is Jain fairness over normalised subscriber entitlements, and  $O_{\text{edge}}$  is measured edge-compute overhead.

This prevents the evaluation from accepting a configuration merely because it improves average throughput while degrading tail latency, fairness, SLA satisfaction, or commodity-edge feasibility.

## 5.6 Ablation Study Plan

The full model is compared with ablated variants:

$$\mathcal{M}_{\text{full}}, \mathcal{M}_{\text{-graph}}, \mathcal{M}_{\text{-policy}}, \mathcal{M}_{\text{-physics}}, \mathcal{M}_{\text{-adaptive}}, \mathcal{M}_{\text{-surv}}, \mathcal{M}_{\text{-LEO}}. \quad (5.4)$$

## 5.7 Evaluation Metrics

**Table 5.2:** Metric-to-research-question alignment.

| RQ    | Metric Class         | Operational Definition                                                                                               |
|-------|----------------------|----------------------------------------------------------------------------------------------------------------------|
| RQ1   | Identity and policy  | Authentication success, EAP/RADIUS latency, realm resolution success, policy-attribute correctness, reject accuracy. |
| RQ2   | Slice enforcement    | Cross-slice leakage, tunnel overhead, per-user bandwidth compliance, shaped latency, accounting mismatch.            |
| RQ2   | Edge feasibility     | CPU, RAM, IRQ, crypto, tunnel, queueing, telemetry overhead.                                                         |
| RQ3   | Optimisation         | Aggregate throughput, fifth-percentile throughput, Jain fairness, SLA violation, interference-cost reduction.        |
| RQ3   | Queue stability      | Estimated $\hat{\sigma}_{k,au}^2$ , finite-buffer loss rate, auxiliary $P(Q_k^\infty > B_k)$ estimate.               |
| RQ1–3 | Deployment viability | Configuration complexity, AP density, terrestrial/LEO sensitivity, NTA compatibility.                                |

## 5.8 Statistical Analysis Plan

For paired comparisons, the treatment effect is

$$\Delta Y = \frac{1}{R} \sum_{r=1}^R (Y_{r,c^*} - Y_{r,c_0}). \quad (5.5)$$

A mixed-effects model accounts for AP, subscriber, and scenario variation:

$$Y_{ijst} = \beta_0 + \beta_1 \text{Config}_i + \beta_2 \text{Density}_j + \beta_3 \text{Backhaul}_k + \beta_4 \text{Mobility}_l + u_s + \varepsilon_{ijst}. \quad (5.6)$$

Because the evaluation compares multiple baselines, scenario classes, and metrics, confirmatory hypothesis tests are corrected using the Holm–Bonferroni family-wise error-rate procedure within each metric family. Exploratory secondary analyses may additionally report Benjamini–Hochberg false-discovery-rate adjusted values, clearly labelled as exploratory. The correction method is fixed before comparative baseline results are analysed.

# Chapter 6

## Implementation Blueprints and Core Algorithms

### 6.1 Authentication and Slice Assignment

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**Algorithm 1:** Safety-Constrained Subscriber Authentication and Slice Assignment

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**Input:** Subscriber request  $Req_i$ , visited AP  $a_j$ , home domain  $d_i$ , policy database  $\mathcal{P}$

**Output:** Access decision, slice assignment, accounting state

- 1 Receive 802.1X/EAPOL association request from device  $s_i$ ;
- 2 Forward EAP payload from  $a_j$  to visited RADIUS proxy;
- 3 Resolve home domain  $d_i$  from subscriber NAI;
- 4 Forward authentication request via RADSEC federation to home RADIUS;
- 5 **if** *EAP-TLS mutual authentication succeeds* **then**
- 6     Retrieve full subscriber policy profile  $p_i$ ;
- 7     Compute  $k^* \leftarrow \text{slice}(p_i, \alpha_i)$ ;
- 8     Verify  $Pol_{ik^*} = 1$ ;
- 9     Evaluate CPU, RAM, IRQ, backhaul, and ring-buffer feasibility for  $k^*$ ;
- 10    **if**  $u = (s_i, a_j, k^*) \in \mathcal{U}_{\text{safe}}(x)$  **then**
- 11     Assign slice  $k^*$  and install VLAN/tc/tunnel rules;
- 12     Initialise RADIUS accounting session  $\rho_i$ ;
- 13     **return** *Access-Accept with slice, QoS, and accounting attributes*;
- 14    **else**
- 15     **return** *Access-Reject with policy/capacity/resource failure reason*;
- 16 **else**
- 17    **return** *Access-Reject with EAP failure reason*;

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## 6.2 Backhaul Selection

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**Algorithm 2:** Safety-First Backhaul Selection

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**Input:** Valid session  $o_i$ , AP state  $a_j$ , terrestrial state  $H_T$ , LEO state  $H_L$ , policy profile  $p_i$

**Output:** Selected backhaul mode and forwarding rule

- 1 Verify pre-condition:  $Auth_i = Pol_i = Iso_i = Acct_i = 1$ ;
- 2 Estimate terrestrial cost  $C_T$  using RTT, jitter, loss, capacity, gateway reachability;
- 3 Estimate LEO cost  $C_L$  using Eq. (4.43);
- 4 **if** *terrestrial path satisfies policy SLA bounds* **then**
- 5     | Select terrestrial path;
- 6 **else**
- 7     | **if** *LEO path is policy-permitted and available* **then**
- 8         | Select LEO path and activate Eq. (3.9);
- 9     | **else**
- 10         | Enter degraded local service if token-cache policy permits;
- 11 Install forwarding rule and record selected path in accounting log;

---

## 6.3 Edge Survivability and Token Cache

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**Algorithm 3:** Cached Token Verification and Offline Accounting

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**Input:** Client request  $Req_i$ , cached token  $\tau_i$ , local key  $K_{net}$ , key age  $\Delta t_{K_{net}}$

**Output:** Offline access decision and signed accounting record

- 1 **if**  $\Delta t_{K_{net}} > 2T_{rot}$  **then**
- 2     | Reject cached mode and apply guest-degraded caps;
- 3 **else**
- 4     | Recompute  $\tau_i^{exp} \leftarrow \text{HMAC-SHA256}_{K_{net}}(ID_i || TS_i || SLA_i)$ ;
- 5     | Verify anti-replay, token equality, and TTL;
- 6     | **if** *all checks pass* **then**
- 7         | Grant capped degraded-mode access:  $b_i^{offline} \leq 0.5b_i^{max}$ ;
- 8         | Append signed offline accounting record
- 9             |  $\hat{\rho}_i \leftarrow \text{HMAC-SHA256}_{K_{net}}(\rho_i || TS_i)$ ;
- 10     | **else**
- 11         | Reject with cached-verification failure;
- 12 On backhaul recovery, flush signed records to home RADIUS and resume online authentication;

---

## 6.4 Algorithmic Complexity and Control Overhead

Algorithm 1 incurs one 802.1X/EAP exchange, one visited-to-home RADIUS/RadSec authentication path, one home-policy lookup, one local edge-feasibility evaluation, and one slice-rule installation per new session. Algorithm 2 evaluates a constant number of candidate backhaul modes per AP and macro epoch, so its online decision overhead is  $O(|\mathcal{H}|)$  per AP. Algorithm 3 performs constant-time HMAC verification per cached-token attempt and appends one signed offline accounting record per accepted degraded-mode session. None of the three algorithms places graph-neural-network inference in the packet fast path; learning-based optimisation remains restricted to the slow macro-policy loop.

# Chapter 7

## Expected Outcomes, Risk Mitigation, and Research Plan

### 7.1 Expected Outcomes

Expected outputs include: a validated subscriber-centric access model; a commodity AP prototype; an edge-resource feasibility dataset; a reproducible emulation and simulation stack; a graph-aware macro-policy optimiser; a terrestrial/LEO backhaul resilience evaluation; and a Nepal-grounded deployment framework.

**Table 7.1:** Research contributions mapped to objectives and questions.

| Contribution | Form         | Description                                                                                                                                                                                          | Addresses       |
|--------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| C1           | Framework    | Subscriber identity as root control variable.                                                                                                                                                        | O1; RQ1         |
| C2           | Architecture | Four-plane architecture with hard safety feasibility.                                                                                                                                                | O3–O4; RQ1–RQ2  |
| C3           | Formal model | Safety-constrained resource allocation combining finite-buffer implementation loss, auxiliary effective-bandwidth overflow estimation, edge-aware graph telemetry, and state-dependent CMDP control. | O4–O5; RQ3      |
| C4           | Methodology  | Typed evidence tracks and ablation protocol.                                                                                                                                                         | O2, O5; RQ2–RQ3 |
| C5           | Deployment   | Nepal scenario classes with optional LEO resilience.                                                                                                                                                 | O6–O8; RQ1–RQ3  |

## 7.2 Validity Strategy

Internal validity is protected through controlled baselines, repeated trials, and isolation of authentication, slicing, tunnelling, queueing, control, and backhaul variables. Construct validity is protected by the metric-to-RQ alignment in Table 5.2. External validity is addressed through four scenario classes rather than a single deployment case. Mathematical validity is protected by separating implementation queues from auxiliary LDT queues, treating safety as state-dependent feasibility, and avoiding unproved root-existence claims.

## 7.3 Reproducibility

All prototype configurations, RADIUS policies, `tc` scripts, topology generators, telemetry schemas, graph model checkpoints, and analysis notebooks will be version-controlled. Experimental outputs will be tagged by evidence class: measured, emulated, simulated, or contextual.

## 7.4 Ethical and Security Considerations

The threat model assumes that the home AAA domain is authoritative for subscriber policy and accounting identity, while visited APs and visited domains are semi-trusted enforcement points. Rogue APs may attempt to impersonate participating infrastructure, replay cached credentials, misapply slice policy, or suppress accounting records. Subscribers may attempt credential sharing, replay, or service-class escalation. A network adversary may observe or disrupt transport links but is not assumed to break standard TLS or HMAC primitives.

The design mitigates these threats by requiring administratively enrolled APs, EAP-TLS or equivalent profile-based subscriber authentication, RadSec-protected inter-domain RADIUS transport, short-lived cached tokens with anti-replay timestamps, key-age rejection after  $2T_{\text{rot}}$ , VLAN/VNI and traffic-control isolation checks, and reconciliation of RADIUS, AP, gateway, and offline signed accounting counters. The cached-token mechanism is treated as an operational survivability mechanism, not as a Byzantine-fault-tolerant security protocol.

Subscriber identities are pseudonymised. No payload inspection is used except on controlled synthetic traffic for isolation testing. Authentication and accounting logs are minimised to evaluation-required fields. Offline cached-token mode uses short token lifetimes, capped bandwidth, anti-replay timestamps, and signed offline accounting. Any field deployment in Nepal will require relevant institutional and regulatory approval.

## 7.5 Risk Register

**Table 7.2:** Reviewer and implementation risk register with mitigation plan.

| Risk                         | Impact                                                         | Mitigation                                                                          |
|------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Commodity AP compute ceiling | Throughput collapses before wireless congestion is measurable. | Treat edge compute as a primary evaluation layer.                                   |
| GNN temporal staleness       | Controller optimises an obsolete graph state.                  | Use two-timescale control and slow macro updates only.                              |
| Queueing formalism overreach | Finite-buffer and LDT models are conflated.                    | Split implementation queue and auxiliary infinite-buffer LDT queue.                 |
| Root existence overclaim     | Effective-bandwidth root may not exist.                        | Use supremum definition; root equality only when crossing conditions hold.          |
| CMDP sign mismatch           | Reward and cost formulations conflict.                         | Use cost minimisation; define reward only as negative cost if needed.               |
| PINN scope creep             | Rain attenuation incorrectly applied to indoor Wi-Fi.          | Use multi-wall Wi-Fi model; reserve ITU-R rain physics for LEO/backhaul.            |
| LEO availability uncertainty | Rural backhaul claim appears speculative.                      | Evaluate LEO as optional resilience, always reporting terrestrial-only baselines.   |
| Safety-constraint bypass     | Security failures treated as performance penalties.            | Use state-dependent safe action set and reject invalid actions before optimisation. |



## 7.6 Work Plan and Timeline

**Table 7.3:** Proposed research timeline with explicit deliverables.

| Phase                     | Period                     | Deliverable                                                             |
|---------------------------|----------------------------|-------------------------------------------------------------------------|
| Mathematical finalisation | July 2026                  | Queueing, LDT, CMDP, GNN, PINN, and LEO equations finalised.            |
| Literature consolidation  | July–August 2026           | Standards map, literature taxonomy, claim-to-citation matrix.           |
| Hardware procurement      | August 2026                | AP cluster, RADIUS server, gateway node, measurement scripts.           |
| Prototype implementation  | September–November 2026    | OpenWrt/FreeRADIUS prototype with VLAN/tc and tunnel slicing.           |
| Baseline experiments      | November–December 2026     | WPA-PSK, captive portal, VLAN-only, greedy, DQN, vanilla GNN baselines. |
| Emulation and simulation  | December 2026–January 2027 | Mininet-WiFi, NS-3, PyTorch Geometric results.                          |
| Nepal scenario validation | February–March 2027        | Four-scenario deployment and regulatory plausibility analysis.          |
| Writing and revision      | March–May 2027             | Submission-ready IEEE TNSM manuscript.                                  |

## 7.7 Limitations

The research is scoped to the access edge. It does not redesign national backbone networks, replace cellular slicing, or guarantee immediate commercial inter-ISP federation. Prototype results on selected OpenWrt-class devices do not generalise to all consumer routers. Learning-based results are conditioned on evidence class and cannot be treated as field-proven unless validated through prototype or deployment evidence. LEO availability, licensing, cost, and handover behaviour vary by region; therefore, LEO is an optional resilience extension, not a core assumption.

## 7.8 Conclusion

This proposal presents a safety-constrained, mathematically corrected, and deployment-aware blueprint for subscriber-centric broadband infrastructure. The revised formulation removes hidden contradictions by separating finite queue implementation from auxiliary LDT analysis, using state-dependent safe action sets, normalising LEO backoff terms, treating HMAC resource targets as empirical criteria, and restricting physics-informed modelling to technically appropriate link types. Nepal functions as the primary empirical testbed because it exposes dense urban Wi-Fi fragmentation, rural mountain backhaul constraints, intermittent infrastructure, and regulatory accountability within one national context.

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